

Ant Colony Optimization for the Traveling Salesman Problem

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Ant Colony Optimization (ACO) is a combinatorial optimization metaheuristic inspired by the foraging behavior of ants. This project implements ACO to solve the Traveling Salesman Problem (TSP) - an NP-hard optimization problem that seeks a minimum-cost Hamiltonian cycle over a set of cities. A population of artificial ants constructs candidate tours probabilistically, guided by pheromone intensity and a distance-based heuristic. After each iteration, pheromone levels are updated to reinforce shorter paths, gradually converging toward a near-optimal solution.

Project goal: Implement a complete ACO solver for TSP with 15–25 randomly generated cities, visualize the search process and best tour in real time using the natID GUI framework, and analyze convergence behavior across iterations. All algorithm parameters (α , β , evaporation rate ρ , number of ants, number of iterations) are kept constant throughout the run to ensure simplicity and stable, reproducible behavior.

Theoretical source: Marco Dorigo & Thomas Stützle, “Ant Colony Optimization” (MIT Press, 2004) - the canonical ACO reference covering the Ant System, ACS, and MMAS variants and their convergence properties.

Starting point / reference implementation: [Professor’s natID example B_S03_Maps](#). The existing Graph, ViewMap (Canvas + animation loop), etc. and background-thread architecture from this example will be adapted and extended. Cities replace towns, TSP tour edges replace roads, and the threaded search loop is reused for the ACO iteration engine.

natID framework usage:

- **Dense matrices (natID):** the $N \times N$ distance matrix D and the $N \times N$ pheromone matrix τ are stored as natID dense matrices. All pheromone update, evaporation, and tour-probability computation steps operate directly on these structures, satisfying the dense/sparse matrix requirement.
- **Optimization problem:** The TSP is formulated as a combinatorial minimization problem over a discrete decision space. The objective function is the total tour length $f(\pi) = \sum d(\pi_i, \pi_{\{i+1\}})$, minimized over all permutations π of N cities, subject to the constraint that each city is visited exactly once. ACO serves as the solver for this problem.
- **natID GUI (Canvas + Animation):** city layout, current best tour, and per-iteration convergence curve are rendered live using natID’s Canvas and animation loop, following the same pattern as B_S03_Maps.

Core development steps:

- City generation and distance matrix construction (natID dense matrix)
- Pheromone matrix initialization and evaporation logic (natID dense matrix operations)
- Probabilistic tour construction per ant using roulette-wheel selection (α , β heuristic) with all parameters fixed as constants
- Global pheromone update: evaporation + deposit proportional to $1/\text{tour_length}$
- Background thread for ACO iteration loop (adapted from B_S03_Maps thread architecture)
- Live GUI: best tour drawn on canvas, convergence curve (cost vs. iteration) as primary output updated each iteration, start/stop/reset controls

Output: The primary visualization is the convergence curve showing best tour cost vs. iteration, updated live each iteration to track solution quality improvement. An animated canvas displays cities, the current best tour, and pheromone strength via edge thickness. A sidebar shows iteration count, best cost found, and runtime.